Solutions by:
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Calculus for Biology and Medicine, 3rd Edition, Claudia Neuhauser

10.2 #2, 16, 18, 23, 29

2) $\lim_{(x,y)\to(-1,1)} (2xy+3x^2) = 2(-1)(1)+3(-1)^2$

$= -2+3 = \boxed{1}$

16) $\lim_{(x,y)\to(0,0)} \frac{3x^2-y^2}{x^2+y^2}$

positive x-axis $\Rightarrow y=0$

$\lim_{x\to0^+} \frac{3x^2-(0)^2}{x^2+(0)^2} = \lim_{x\to0^+} \frac{3x^2}{x^2}$

$= \lim_{x\to0^+} 3$

$= 3$

positive y-axis $\Rightarrow x=0$

$\lim_{y\to0^+} \frac{3(0)^2-y^2}{(0)^2+y^2} = \lim_{y\to0^+} \frac{-y^2}{y^2}$

$= \lim_{y\to0^+} -1$

$= -1$

$\boxed{3 \neq -1 \Rightarrow \lim_{(x,y)\to(0,0)} \frac{3x^2-y^2}{x^2+y^2} \text{ DNE}}$

18) $\lim_{(x,y)\to(0,0)} \frac{3xy}{x^2+y^3}$

along $y=mx$

$\lim_{x\to0} \frac{3x(mx)}{x^2+(mx)^3} = \lim_{x\to0} \frac{x^2}{x^2}\frac{3m}{1+m^3x}$

$= \lim_{x\to0} \frac{3m}{1+m^3x}$

$= \frac{3m}{1+m^3(0)}$

$= 3m$

Since the limit is not constant and depends on the direction, the limit does not exist.

23) $f(x,y) = \begin{cases} \dfrac{4xy}{x^2+y^2} & \text{for } (x,y) \neq (0,0) \\ 0 & \text{for } (x,y) = (0,0) \end{cases}$

$$\lim_{(x,y)\to(0,0)} f(x,y) = \lim_{(x,y)\to(0,0)} \frac{4xy}{x^2+y^2}$$

along x-axis $\Rightarrow y=0$

$$\lim_{x\to 0} \frac{4x(0)}{x^2+(0)^2} = \lim_{x\to 0} 0 = 0$$

along y-axis $\Rightarrow x=0$

$$\lim_{y\to 0} \frac{4(0)y}{(0)^2+y^2} = \lim_{y\to 0} 0 = 0$$

along $y=x$

$$\lim_{x\to 0} \frac{4x(x)}{x^2+(x)^2} = \lim_{x\to 0} \frac{4x^2}{2x^2} = \lim_{x\to 0} 2 = 2$$

$0 \neq 2 \Rightarrow \lim_{(x,y)\to(0,0)} f(x,y)$ DNE

Therefore, $f(x,y)$ is not continuous at $(0,0)$.

29) $h(x,y) = e^{xy}$

(a) $\boxed{f(x,y) = xy,\ g(z) = e^z}$

$h(x,y) = (g\circ f)(x,y) = e^{xy}$

(b) $f(x,y)$ is continuous $\forall (x,y) \in \mathbb{R}^2$, and $g(z)$ is continuous $\forall z \in \mathbb{R}$. Therefore, $h(x,y)$ is continuous $\forall (x,y) \in \mathbb{R}^2$